Proof:

If $P(B) > 0$, multiplying both sides of $P(A/B) = \frac{P(A \cap B)}{P(B)}$ by $P(B)$, we get

$P(A \cap B) = P(B)\, P(A/B)$

If $P(A) > 0$, interchanging A and B in $P(A/B) = \frac{P(A \cap B)}{P(B)}$, it also follows that

$P(A \cap B) = P(A)P(B/A)$.

Hence, the theorem is proved.



NOTE

If A and B are any two events of an experiment, then $P(A \cap B)$ can be calculated using the conditional probability of A given B or of B given A. Multiplication theorem of probability can be extended to compute $P(A \cap B \cap C)$ for the events A, B and C as follows:

$$P(A \cap B \cap C) = P(A)P(B/A)P(C/A \cap B).$$

Example 8.18

A box contains 7 red and 3 white marbles. Three marbles are drawn from the box one after the other without replacement. Find the probability of drawing three marbles in the alternate colours with the first marble being red.

Solution:

The event of interest is drawing the marbles in alternate colours with the first is red. This event can occur only when the marbles are drawn in the order (Red , White , Red)

If A and C represent the events of drawing red marbles respectively in the first and the third draws and B is the event of drawing white marble in the second draw, then the required event is $A \cap B \cap C$. The probability for the occurrence of $A \cap B \cap C$ can be calculated applying

$$P(A \cap B \cap C) = P(A)P(B/A)P(C/A \cap B)$$

Since there are 7 red and 3 white marbles in the box for the first draw,

$$P(A) = \frac{7}{10}$$

Now, there will be 6 red and 3 white marbles in the box for the second draw if the event A has occurred. Hence,

$$P(B/A) = \frac{3}{9}$$